

# A Fault Taxonomy for the Apollo LM Landing Problem

Study G — landing probability across 16 engine faults and 5 initial-condition regimes

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## 1 What this study adds

The earlier case studies each took **one** fault and flew it as well as it can be flown: an engine out (Study A), a thrust-efficiency sweep (Study B), a gimbal bandwidth sweep (Study C), fault onset time (Studies D/E), and whether a fault is expressible as an initial condition (Study F, which found that it is not — the state is recoverable information, the *plant change* is not).

Between them they cover three of the eleven dynamic-effect categories in `docs/spacecraft_engine_fault_framework.md`. This study covers the rest. It instantiates **15 distinct faults** drawn from that framework — one per dynamic-effect category, with severity variants where the category spans a wide range — as 15 *plants*, and measures the landing probability of each from **5 qualitatively different regions of the state space**.

The two axes are the ones the framework itself insists on:

- **What the fault is.** Framework section 3 classifies every fault as *additive* (an unknown disturbance, independent of state and input), *multiplicative* (a change to  $A$  or  $B$ , scaling with the operating point), or *structural* (a change to the model's order or topology). This is the classification that decides what a fault-tolerant architecture has to *do*, so it is the one the figures colour by.

- **Where the vehicle is when it gets it.** A fault that is survivable on a high, calm approach is not the same fault at 300 m with the vehicle already rolling. Reporting a single landing rate per fault would average those two situations into a number describing neither.

The deliverable is therefore the landing probability over the **product** of the two: 16 faults  $\times$  5 regimes  $\times$  10 initial conditions = **800 nonlinear programmes** (15.6 core-hours of IPOPT).

## 1.1 The experiment, in one paragraph

Initial conditions are drawn once per regime, as scrambled Sobol points in the same 22-dimensional box Study F used (12 rigid-body states plus all 10 actuator states), and **reused for every fault**. Each sample is then handed to the recovery planner of `fault_lib.recover`: given the state the vehicle is in and the plant it has become, plan a landing to the 80 s deadline or refuse. There is no nominal trajectory and no fault onset time — Study F established that the post-fault problem is fully specified by (state, plant), and the fault enters here only through the plant, which is precisely the part that is *not* an initial condition. The resulting trajectory is flown to the contact altitude, the engine is cut, the vehicle settles ballistically, and the touchdown state is scored against the Apollo landing gate (vertical speed  $\leq 3.0$  m/s, horizontal  $\leq 1.2$  m/s, tilt  $\leq 6^\circ$ , 15 m of the pad, rates  $\leq 5^\circ/\text{s}$ ).

Because the initial conditions are shared across plants, the healthy configuration is a genuine control: any difference within a regime column is attributable to the plant alone, and the healthy-vs-fault comparison is a within-sample McNemar test rather than two independent proportions.

## 2 The fault catalogue

Every fault acts on engine 2; engine 1 stays healthy, so each case is an *asymmetry* the vehicle must trim as well as a loss of performance. Severity is chosen to span the interesting range rather than the survivable one — a taxonomy in which everything lands measures nothing.

| Fault                                 | §   | FTC structure  | Temporal              | What the plant becomes                                                                                                          |
|---------------------------------------|-----|----------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Healthy (control)                     | —   | none           | —                     | nominal plant; measures what the state dispersion alone costs                                                                   |
| Thrust-vector misalignment (3°, 2°)   | 2.7 | additive       | incipient             | asymmetric nozzle erosion leaves engine 2 pointing 3° in pitch and 2° in yaw off its commanded axis, independent of command     |
| Thrust oscillation (chugging)         | 2.3 | additive       | intermittent / forced | feed-coupled instability: engine 2 thrust rings at 0.25 Hz with an amplitude of 20 % of its hover share, regardless of throttle |
| Thrust reduction, $\eta=0.50$         | 2.1 | multiplicative | abrupt / incipient    | engine 2 delivers half of its internal thrust (turbopump degradation, injector blockage, throat erosion)                        |
| Thrust reduction, $\eta=0.15$         | 2.1 | multiplicative | abrupt                | engine 2 delivers 15 % — near-total gain loss with the gimbal alive                                                             |
| Thrust excess, $\eta=1.30$            | 2.2 | multiplicative | abrupt                | pressurant regulator runaway: engine 2 delivers 130 % of its commanded thrust                                                   |
| Slow thrust response, $\tau_T=2.5$ s  | 2.5 | multiplicative | incipient             | engine 2 thrust lag grows from 0.4 s to 2.5 s (valve friction, coking, actuator supply loss) with the gain untouched            |
| Gimbal bandwidth loss, $\omega_n=0.6$ | 2.5 | multiplicative | incipient             | engine 2 gimbal actuator slowed from 4.0 to 0.6 rad/s                                                                           |

| Fault                            | §          | FTC structure               | Temporal              | What the plant becomes                                                                                                                                                                                              |
|----------------------------------|------------|-----------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gimbal underdamped, $\zeta=0.25$ | 2.3<br>/   | multiplicative              | incipient             | engine 2 gimbal at $\omega_n=1.0$ rad/s with damping down to 0.25 — a lightly damped TVC mode                                                                                                                       |
| TVC effectiveness loss (35 %)    | 2.8<br>2.7 | multiplicative              | incipient             | engine 2 gimbal reaches only 35 % of the commanded deflection                                                                                                                                                       |
| Mixture-ratio shift (coupled)    | 2.4        | multiplicative (coupled)    | incipient             | oxidiser-side erosion on engine 2: gain falls to 0.75 <i>and</i> the combustion time constant grows to 1.2 s — the framework’s coupled $\Delta B + \Delta A$ case, where the gain change drags the dynamics with it |
| Throat-erosion drift             | 2.10       | time-varying multiplicative | incipient             | engine 2 efficiency decays at 1.2 %/s from the moment the planner starts: healthy at $t=0$ , $\eta=0.52$ forty seconds later. The vehicle it plans with is not the vehicle it lands with                            |
| Valve stuck open (thrust floor)  | 2.2<br>/   | structural                  | abrupt                | engine 2 cannot be throttled below 8458 N (1.35x its hover share) — the input set itself is cut                                                                                                                     |
| Engine out                       | 2.1<br>/   | structural                  | abrupt                | engine 2 dead: thrust and both gimbals pinned to zero for the rest of the flight                                                                                                                                    |
| Gimbal seizure                   | 3.1<br>1.7 | structural                  | abrupt                | engine 2 gimbal bearing seizes: the deflection freezes at whatever it held at the fault and no longer answers the command                                                                                           |
| Transport delay (1 interval)     | 2.6        | structural                  | abrupt / intermittent | a vapour pocket in engine 2’s feed line delays every command by one full control interval (1 s) — exact on a zero-order-hold grid, so no Pade approximation is involved                                             |

Four of these required extending the vehicle model, because the earlier studies only ever needed thrust efficiency and gimbal bandwidth. The additions are all in `LMPParams` and `flat_moon_6dof`, and all default to the healthy plant:

- **transport delay** is exact rather than a Padé approximation — on a zero-order-hold control grid, a delay of one interval is a pure index shift of the engine’s command, which is what the OCP builder now applies;
- **parametric drift** and **forced oscillation** required threading trajectory time through the integrator, so a time-varying plant can be integrated at all;
- **thrust-vector misalignment** enters as a bias on the delivered thrust axis that no command can null — the planner can only aim the sum somewhere useful;
- **valve stuck open** is not a parameter change but a cut to the *input set*: the engine’s lower thrust bound is raised above its own hover share;
- **gimbal seizure** freezes the deflection where the fault caught it. The seizure arrests the motion, so the gimbal *rate* states are zeroed in the initial condition and the acceleration is identically zero thereafter. A first attempt instead bled the rate off with a fast decay term, which looks more physical and is numerically wrong: a decay fast enough to read as a seizure sits far outside RK4’s stability region on this grid, so the rate state diverged and every seizure solve came back infeasible for reasons that had nothing to do with the vehicle.

### 3 The initial-condition regimes

The regimes were **calibrated rather than guessed**. A pilot sweep established that this recovery planner, handed the damaged plant exactly and a free choice of trajectory, absorbs almost every single-engine fault from a high, calm approach — the healthy control and most of the catalogue all land, and the matrix saturates at 1.00. Regime boxes were therefore tightened until the *healthy* vehicle itself starts failing, which is the only condition under which the fault differences are measurable at all. That saturation is itself a result, and it is reported below rather than tuned away: the first three regimes are kept precisely because they show where the ceiling is.

| Regime       | What it is                                                                                                                                                                                                  | Landing<br>rate,<br>all<br>faults<br>pooled | n       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------|
| On approach  | high, descending, small dispersions — the vehicle is where a healthy descent would have put it, and the fault is the only thing wrong                                                                       | <b>0.88</b><br>[0.81,<br>0.92]              | 140/160 |
| Dispersed    | Study F's wide box: position, velocity, attitude, rates and both sets of actuator states drawn over their full admissible ranges                                                                            | <b>0.78</b><br>[0.70,<br>0.83]              | 124/160 |
| Upset        | the corner of the box where the vehicle is already rotating or tilted ( $\geq 10$ deg/s on some axis, or $\geq 15$ deg of tilt) with the gimbals deflected — the states a fault transient actually produces | <b>0.75</b><br>[0.68,<br>0.81]              | 120/160 |
| Low and late | between 60 m and 200 m and still descending: the fault arrives when there is very little altitude left to trade for time                                                                                    | <b>0.66</b><br>[0.58,<br>0.73]              | 105/160 |
| Critical     | below 140 m, descending hard, tilted and rotating — the regime where even the healthy vehicle frequently has no trajectory left, so the fault has to be paid for out of a margin that is already spent      | <b>0.41</b><br>[0.34,<br>0.49]              | 66/160  |

The regimes are **not nested severity levels**; they name qualitatively different situations. **On approach** is the most forgiving (0.88 pooled over every fault) and **Critical** the least (0.41) — a spread of 0.46 in landing probability produced *entirely* by where the vehicle started, with the fault mix held identical.

## 4 Results

### 4.1 The fault $\times$ regime matrix

This is the study's primary result: the landing probability of every fault from every regime, over shared initial conditions.

| Landing probability by fault and initial-condition regime<br>10 shared initial conditions per cell; label colour = FTC fault structure |                                                                                     |           |       |              |          |               |
|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------|-------|--------------|----------|---------------|
| Healthy (control)                                                                                                                      | 1.00                                                                                | 1.00      | 1.00  | 0.80         | 0.60     | 0.88          |
| Thrust-vector misalignment ( $3^\circ, 2^\circ$ )                                                                                      | 1.00                                                                                | 1.00      | 1.00  | 0.80         | 0.60     | 0.88          |
| Thrust oscillation (chugging)                                                                                                          | 1.00                                                                                | 1.00      | 1.00  | 0.80         | 0.60     | 0.88          |
| Thrust reduction, $\eta=0.50$                                                                                                          | 1.00                                                                                | 0.90      | 0.70  | 0.50         | 0.20     | 0.66          |
| Thrust reduction, $\eta=0.15$                                                                                                          | 0.00                                                                                | 0.00      | 0.00  | 0.00         | 0.00     | 0.00          |
| Thrust excess, $\eta=1.30$                                                                                                             | 1.00                                                                                | 1.00      | 1.00  | 0.80         | 0.60     | 0.88          |
| Slow thrust response, $\tau_T=2.5$ s                                                                                                   | 1.00                                                                                | 1.00      | 0.80  | 0.70         | 0.30     | 0.76          |
| Gimbal bandwidth loss, $\omega_n=0.6$                                                                                                  | 1.00                                                                                | 0.40      | 0.40  | 0.70         | 0.30     | 0.56          |
| Gimbal underdamped, $\zeta=0.25$                                                                                                       | 1.00                                                                                | 0.80      | 0.80  | 0.80         | 0.50     | 0.78          |
| TVC effectiveness loss (35 %)                                                                                                          | 1.00                                                                                | 1.00      | 1.00  | 0.80         | 0.60     | 0.88          |
| Mixture-ratio shift (coupled)                                                                                                          | 1.00                                                                                | 0.90      | 0.90  | 0.70         | 0.30     | 0.76          |
| Throat-erosion drift                                                                                                                   | 1.00                                                                                | 0.80      | 0.80  | 0.80         | 0.60     | 0.80          |
| Valve stuck open (thrust floor)                                                                                                        | 1.00                                                                                | 0.80      | 0.80  | 0.80         | 0.60     | 0.80          |
| Engine out                                                                                                                             | 0.00                                                                                | 0.00      | 0.00  | 0.00         | 0.00     | 0.00          |
| Gimbal seizure                                                                                                                         | 1.00                                                                                | 0.90      | 0.90  | 0.80         | 0.50     | 0.82          |
| Transport delay (1 interval)                                                                                                           | 1.00                                                                                | 0.90      | 0.90  | 0.70         | 0.30     | 0.76          |
|                                                                                                                                        | On approach                                                                         | Dispersed | Upset | Low and late | Critical | all<br>(n=50) |
|                                                                                                                                        | <div> <div></div> additive <div></div> multiplicative <div></div> structural </div> |           |       |              |          |               |

Figure 1: Landing probability by fault and initial-condition regime

Read down a column to compare faults at a fixed situation; read across a row to see how much of a given fault's cost is really the situation's. Each cell rests on 10 samples, so a single cell carries a 95 % interval roughly  $\pm\$0.25$  wide — the cells are for the *pattern*, and every number quoted in the text below is a pooled marginal with its interval attached.

## 4.2 What each fault costs, pooled

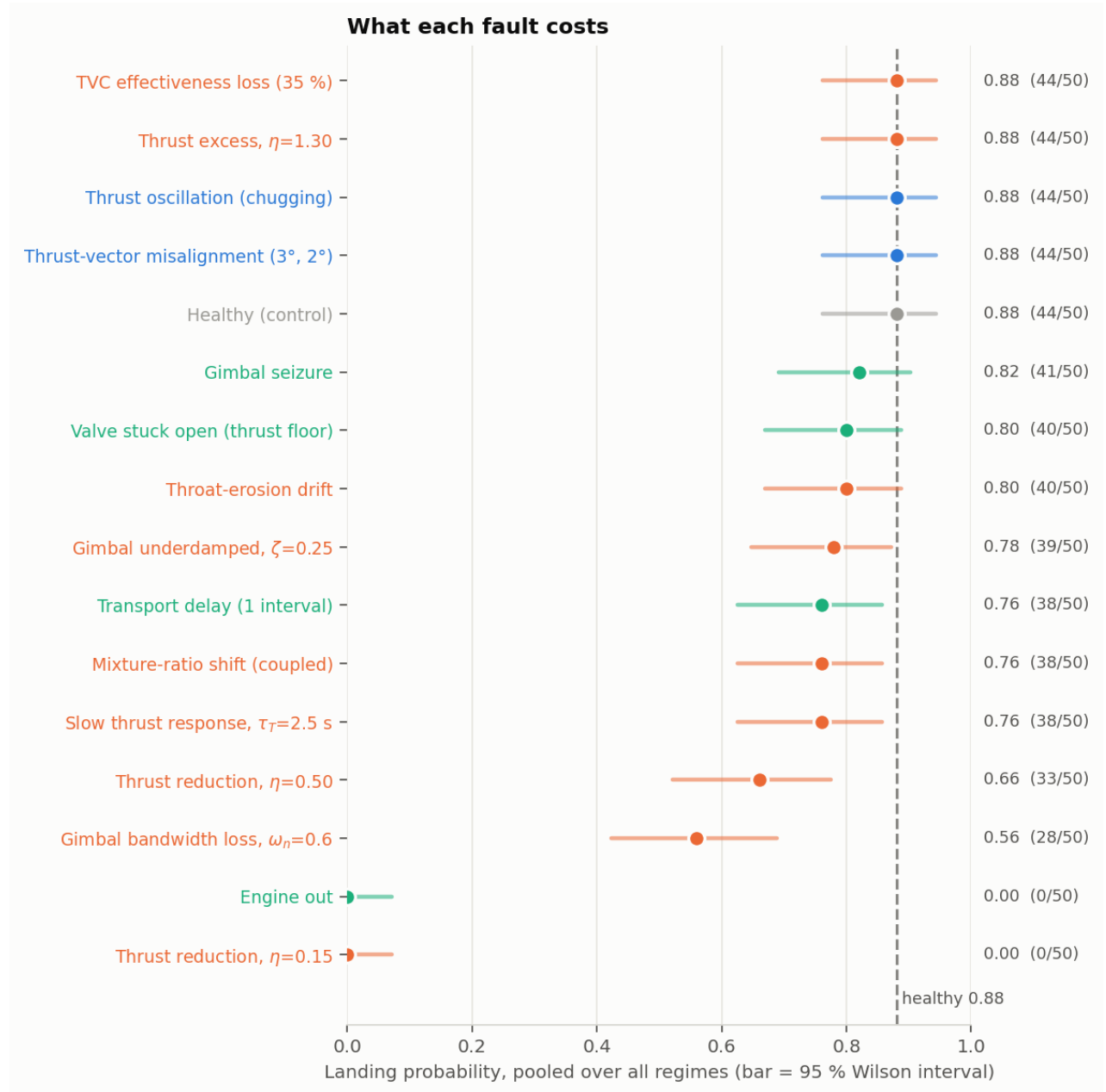


Figure 2: Pooled landing rate per fault, Wilson intervals

| Fault                                                | Class          | Landing rate      | k/n   | Survival | Lost | McNemar     |
|------------------------------------------------------|----------------|-------------------|-------|----------|------|-------------|
| Healthy (control)                                    | none           | 0.88 [0.76, 0.94] | 44/50 | —        | —    | —           |
| Thrust-vector misalignment ( $3^\circ$ , $2^\circ$ ) | additive       | 0.88 [0.76, 0.94] | 44/50 | 1.00     | 0    | $p = 1.000$ |
| Thrust oscillation (chugging)                        | additive       | 0.88 [0.76, 0.94] | 44/50 | 1.00     | 0    | $p = 1.000$ |
| Thrust excess, $\eta=1.30$                           | multiplicative | 0.88 [0.76, 0.94] | 44/50 | 1.00     | 0    | $p = 1.000$ |

| Fault                                 | Class          | Landing rate      | k/n   | Survival | Lost | McNemar       |
|---------------------------------------|----------------|-------------------|-------|----------|------|---------------|
| TVC effectiveness loss (35 %)         | multiplicative | 0.88 [0.76, 0.94] | 44/50 | 1.00     | 0    | $p = 1.000$   |
| Gimbal seizure                        | structural     | 0.82 [0.69, 0.90] | 41/50 | 0.93     | 3    | $p = 0.250$   |
| Throat-erosion drift                  | multiplicative | 0.80 [0.67, 0.89] | 40/50 | 0.91     | 4    | $p = 0.125$   |
| Valve stuck open (thrust floor)       | structural     | 0.80 [0.67, 0.89] | 40/50 | 0.91     | 4    | $p = 0.125$   |
| Gimbal underdamped, $\zeta=0.25$      | multiplicative | 0.78 [0.65, 0.87] | 39/50 | 0.89     | 5    | $p = 0.062$   |
| Slow thrust response, $\tau_T=2.5$ s  | multiplicative | 0.76 [0.63, 0.86] | 38/50 | 0.86     | 6    | $p = 0.031$   |
| Mixture-ratio shift (coupled)         | multiplicative | 0.76 [0.63, 0.86] | 38/50 | 0.86     | 6    | $p = 0.031$   |
| Transport delay (1 interval)          | structural     | 0.76 [0.63, 0.86] | 38/50 | 0.86     | 6    | $p = 0.031$   |
| Thrust reduction, $\eta=0.50$         | multiplicative | 0.66 [0.52, 0.78] | 33/50 | 0.75     | 11   | $p = 0.001$   |
| Gimbal bandwidth loss, $\omega_n=0.6$ | multiplicative | 0.56 [0.42, 0.69] | 28/50 | 0.64     | 16   | $p < 10^{-4}$ |
| Thrust reduction, $\eta=0.15$         | multiplicative | 0.00 [0.00, 0.07] | 0/50  | 0.00     | 44   | $p < 10^{-4}$ |
| Engine out                            | structural     | 0.00 [0.00, 0.07] | 0/50  | 0.00     | 44   | $p < 10^{-4}$ |

Landing rate carries its 95 % Wilson interval; *survival* is the paired share of healthy-landable states the fault keeps, *lost* the count it takes away, and *McNemar* the exact two-sided test on the discordant pairs.

The healthy control lands from **0.88** [0.76, 0.94] of the sampled states. That number is not 1.00 and is not meant to be: the box contains states from which a *healthy* Apollo LM cannot reach the pad inside the deadline, and pricing them is the whole reason the control is run on the same samples.

The most expensive fault in the catalogue is **Thrust reduction,  $\eta=0.15$  (0.00 [0.00, 0.07])**; the cheapest is **Thrust-vector misalignment (3°, 2°) (0.88 [0.76, 0.94])**. 7 of the 15 faults degrade the landing rate significantly against the paired healthy control at the 5 % level.

The engine-out row is a **cross-check on the whole apparatus rather than a new result**: Study A’s roll-authority budget says a single gimbal can trim a one-engine-out asymmetry only if  $y_{eng} \leq dz_{eng} \tan \delta_{max} = 0.263$  m, and this vehicle is built at  $y_{eng} = 1.5$  m. It should therefore never land engine-out at any initial condition, and it does not (0/50). A campaign that produced engine-out landings here would be reporting a bug.

### 4.3 Paired survival: the same states, a different vehicle

The pooled rate above still mixes the fault’s cost with the cost of the state dispersion. The paired view removes the latter: of the initial conditions the *healthy* vehicle lands from, what share does the faulted vehicle still land from? The initial conditions are literally identical, so what is left is the plant.

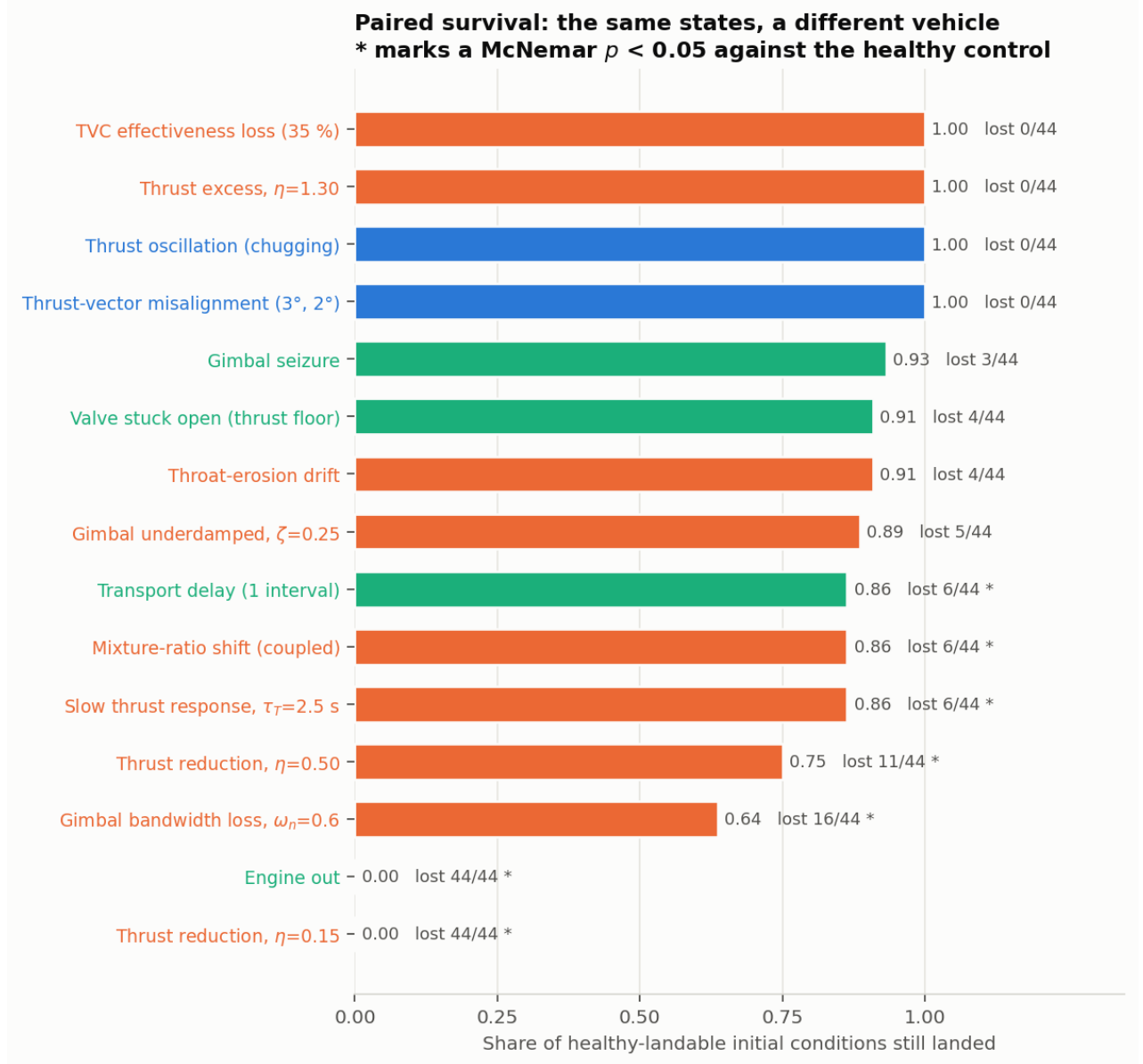


Figure 3: Paired survival relative to the healthy control

11 faults keep at least 85 % of the healthy vehicle’s landable states (Thrust-vector misalignment ( $3^\circ$ ,  $2^\circ$ ), Thrust oscillation (chugging), Thrust excess,  $\eta=1.30$ , Slow thrust response,  $\tau_T=2.5$  s, Gimbal underdamped,  $\zeta=0.25$ , TVC effectiveness loss (35 %), Mixture-ratio shift (coupled), Throat-erosion drift, Valve stuck open (thrust floor), Gimbal seizure, Transport delay (1 interval)). These are faults the *planner* absorbs: it knows the damaged plant, and it re-optimises around it. That is a statement about open-loop replanning with a perfect plant estimate, not about a controller that has to discover the fault first — see the limitations.

At the other end, 2 faults destroy at least 65 % of them (Thrust reduction,  $\eta=0.15$ , Engine out). No amount of replanning recovers these, because the vehicle no longer has the authority the trajectory requires.

#### 4.3.1 The additive faults cost nothing at all

Every additive fault in the catalogue (Thrust-vector misalignment ( $3^\circ$ ,  $2^\circ$ ), Thrust oscillation (chugging)) loses **zero** of the healthy vehicle’s landable initial conditions — not few, zero. This is the framework’s section 4



prediction landing exactly: an additive fault is an unknown input that does not depend on state or command, and an optimiser that knows the disturbance simply builds it into the plan. The corollary is the part that matters operationally: their cost is *entirely* a detection and estimation problem. Give the planner the wrong bias and it aims the trajectory wrong; give it the right one and the fault is free.

#### 4.3.2 A sluggish gimbal is worse than a seized one

The catalogue’s most counter-intuitive result, and it is not a small margin: the seized gimbal keeps 0.93 of the healthy vehicle’s landable states while the merely *slow* one keeps 0.64 — pooled landing rates 0.82 against 0.56.

The reason is that the two faults present the planner with different kinds of problem. A seized gimbal is a **constant, known offset**: the deflection is frozen where the fault caught it, the resulting force and moment are fixed for the rest of the flight, and the trajectory can be planned around a constant. A slow gimbal is a **lagging actuator that still moves**: it answers every command late and in the wrong amount during exactly the transients where the authority is needed, and the planner cannot get the deflection where it wants it when it wants it. Losing an actuator cleanly is easier to fly than keeping one that is late — which is a direct argument for the framework’s section 7 detection layer being worth more than raw actuator redundancy, because a fault you have identified as a fixed offset has largely stopped being a fault.

### 4.4 How the failures fail

Landing rate alone hides the engineering distinction between a vehicle that *arrives too hard* and a vehicle for which no arrival exists at all. The first is a trajectory-shaping problem; the second is a control-authority problem, and no better guidance law fixes it.

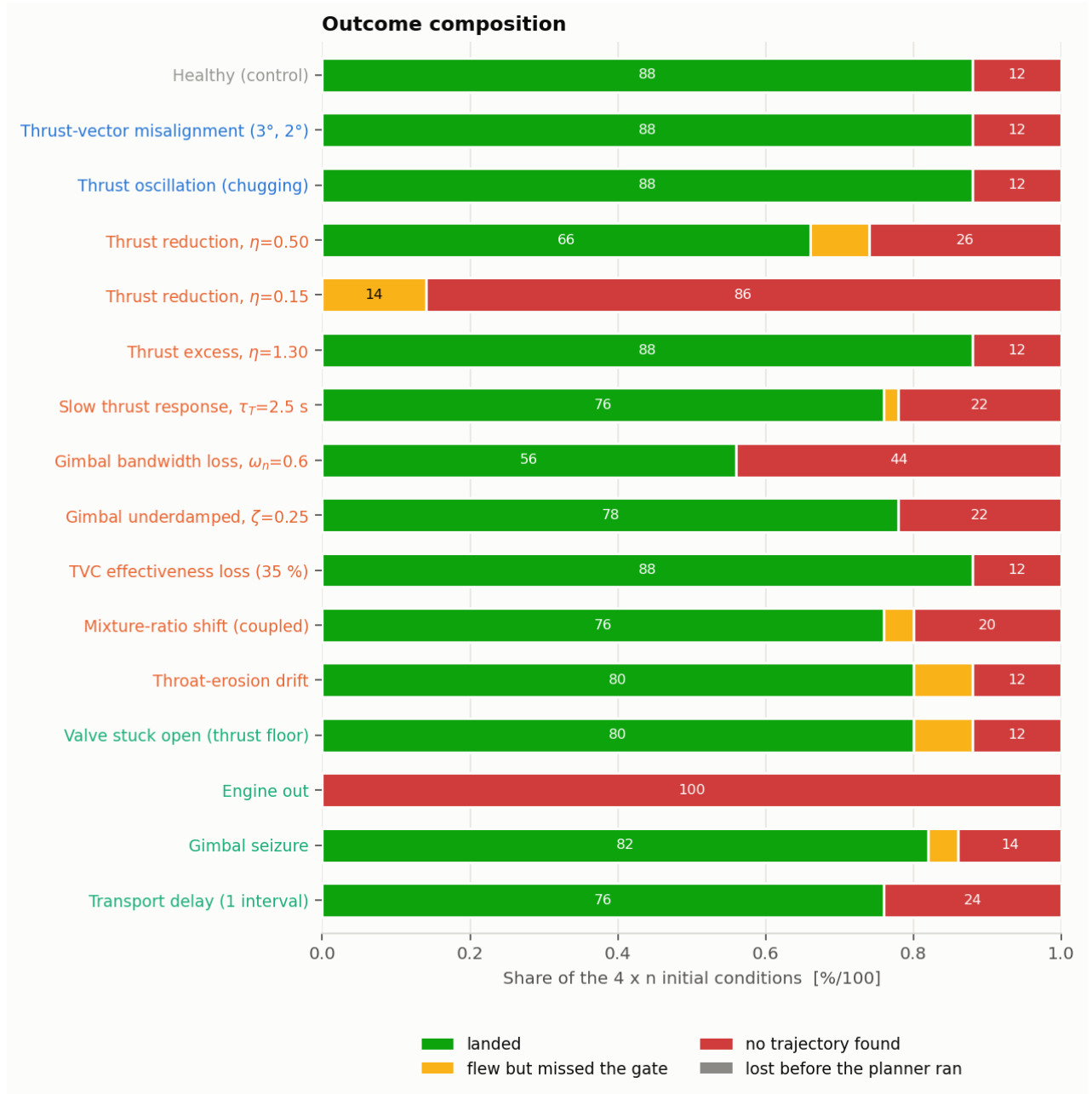


Figure 4: Outcome composition per fault

| Fault                                             | landed | gate miss | no trajectory | lost first |
|---------------------------------------------------|--------|-----------|---------------|------------|
| Healthy (control)                                 | 88 %   | 0 %       | 12 %          | 0 %        |
| Thrust-vector misalignment ( $3^\circ, 2^\circ$ ) | 88 %   | 0 %       | 12 %          | 0 %        |
| Thrust oscillation (chugging)                     | 88 %   | 0 %       | 12 %          | 0 %        |
| Thrust reduction, $\eta=0.50$                     | 66 %   | 8 %       | 26 %          | 0 %        |
| Thrust reduction, $\eta=0.15$                     | 0 %    | 14 %      | 86 %          | 0 %        |
| Thrust excess, $\eta=1.30$                        | 88 %   | 0 %       | 12 %          | 0 %        |
| Slow thrust response, $\tau_T=2.5$ s              | 76 %   | 2 %       | 22 %          | 0 %        |
| Gimbal bandwidth loss, $\omega_n=0.6$             | 56 %   | 0 %       | 44 %          | 0 %        |
| Gimbal underdamped, $\zeta=0.25$                  | 78 %   | 0 %       | 22 %          | 0 %        |

| Fault                           | landed | gate miss | no trajectory | lost first |
|---------------------------------|--------|-----------|---------------|------------|
| TVC effectiveness loss (35 %)   | 88 %   | 0 %       | 12 %          | 0 %        |
| Mixture-ratio shift (coupled)   | 76 %   | 4 %       | 20 %          | 0 %        |
| Throat-erosion drift            | 80 %   | 8 %       | 12 %          | 0 %        |
| Valve stuck open (thrust floor) | 80 %   | 8 %       | 12 %          | 0 %        |
| Engine out                      | 0 %    | 0 %       | 100 %         | 0 %        |
| Gimbal seizure                  | 82 %   | 4 %       | 14 %          | 0 %        |
| Transport delay (1 interval)    | 76 %   | 0 %       | 24 %          | 0 %        |

**lost first** is the share of samples where the initial state is already past the hard loss-of-control criteria (tilt beyond  $90^\circ$ , tumbling beyond  $120^\circ/\text{s}$ , speed beyond  $1.5 V_{max}$ , or below the contact altitude); it is a property of the *sample*, not of the fault, and is therefore identical across every row — a useful internal consistency check on the pairing.

#### 4.5 Landing probability against the initial condition

The regimes are boxes. This is the continuous cut through them: landing rate against initial altitude, with the fault classes separated.

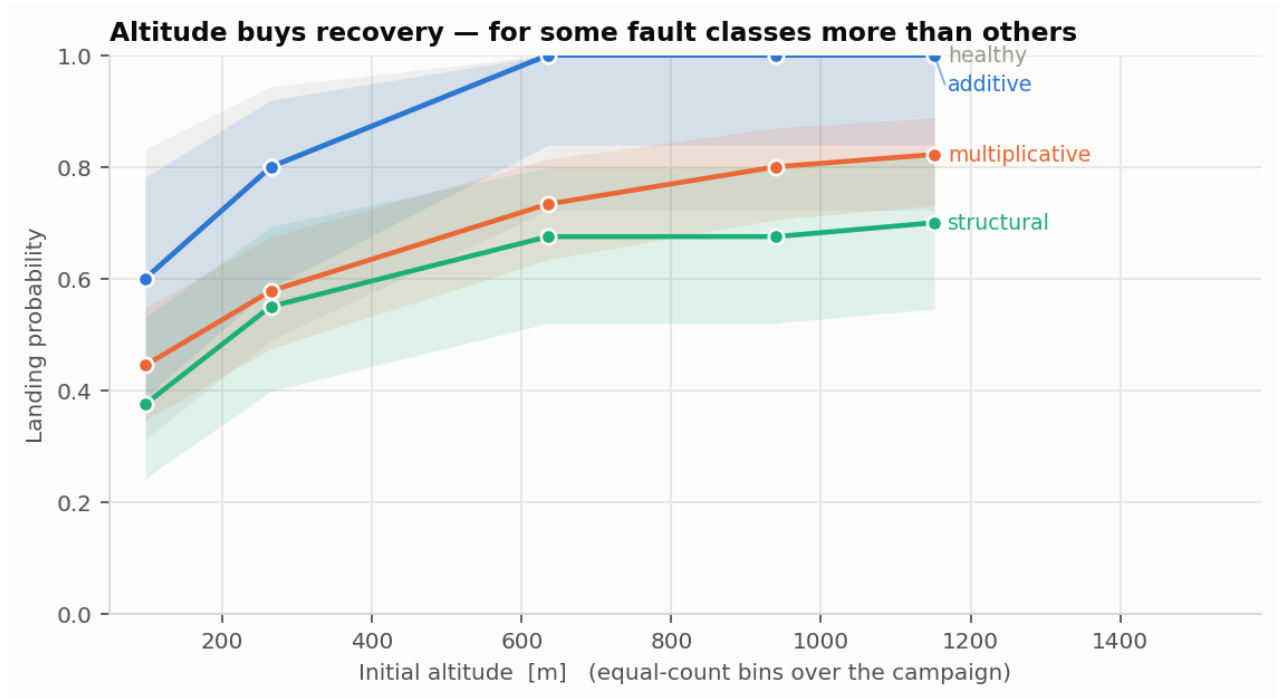


Figure 5: Landing rate against initial altitude by class

| Fault class    | On approach | Dispersed | Upset | Low and late | Critical | pooled                   |
|----------------|-------------|-----------|-------|--------------|----------|--------------------------|
| healthy        | 1.00        | 1.00      | 1.00  | 0.80         | 0.60     | <b>0.88</b> [0.76, 0.94] |
| additive       | 1.00        | 1.00      | 1.00  | 0.80         | 0.60     | <b>0.88</b> [0.80, 0.93] |
| multiplicative | 0.89        | 0.76      | 0.71  | 0.64         | 0.38     | <b>0.68</b> [0.63, 0.72] |

| Fault class | On approach | Dispersed | Upset | Low and late | Critical | pooled                   |
|-------------|-------------|-----------|-------|--------------|----------|--------------------------|
| structural  | 0.75        | 0.65      | 0.65  | 0.57         | 0.35     | <b>0.59</b> [0.53, 0.66] |

#### 4.6 How close the survivors came

Landing is a binary, but the gate margin behind it is not: a margin of 1.0 *is* the gate, and the distance below it says how much of the touchdown budget the recovery had to spend.

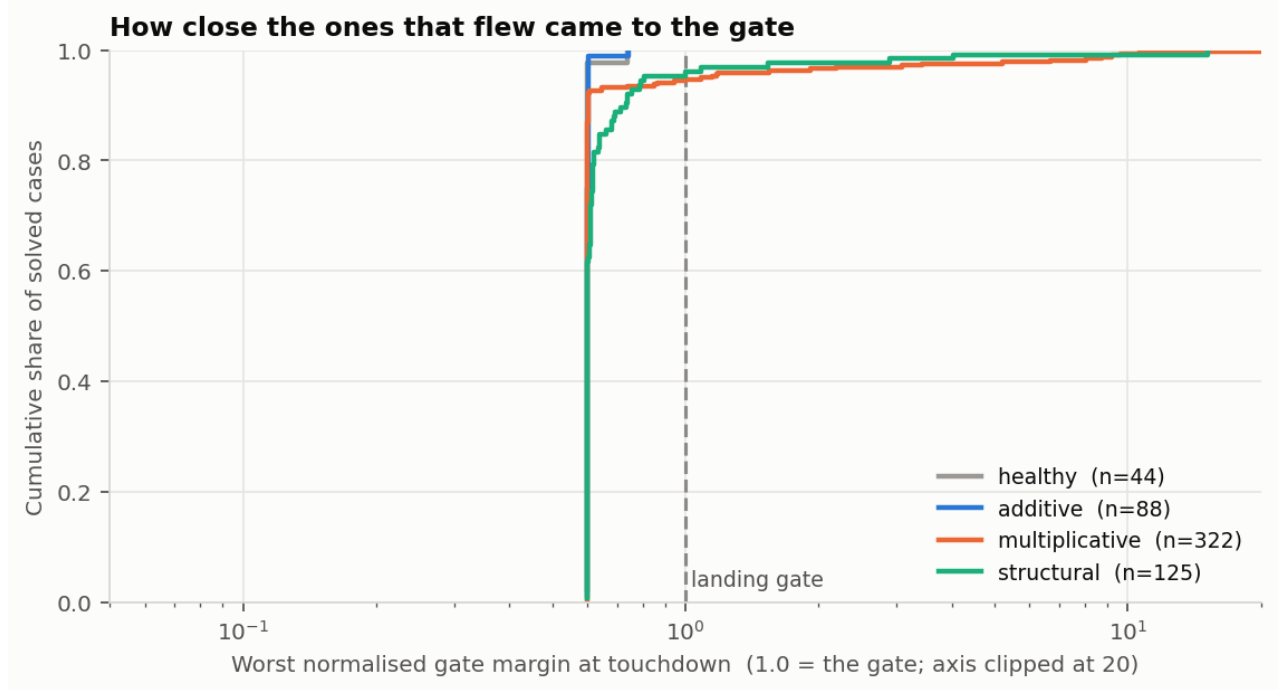


Figure 6: Gate-margin distribution over solved cases

### 5 What this means for a fault-tolerant architecture

**The framework’s three-way structural classification predicts the outcome better than fault severity does.** Grouped by class, the pooled landing rates are healthy 0.88, additive 0.88, multiplicative 0.68, structural 0.59. Structural faults are the ones that remove authority outright; multiplicative faults scale with the operating point, so the planner can trade against them; additive faults are biases and disturbances that an optimiser with a correct plant model simply plans around.

**Where the vehicle is matters as much as what broke.** The regime spread (0.41 to 0.88) is comparable to the spread across the fault catalogue itself. A fault-tolerance budget quoted as a single per-fault probability is not a budget; it is an average over a mission phase distribution that was never stated.

**Altitude is the resource.** The **Critical** regime differs from the others mainly in having less of it, and it is where the fault classes separate most sharply. Recovery is paid for in altitude: the same fault that is a nuisance on approach, where there is a kilometre of it to spend, is what decides the landing once the vehicle is low.

**The three FTC layers of framework section 7 are visible in the data.** The faults the planner absorbs are the ones a *base* robust/adaptive layer would handle; the ones that produce **gate miss** rather than **no trajectory** are disturbance-estimation problems; the ones that produce **no trajectory** need the detection-and-switching layer, and in several cases need an abort decision rather than a control law.

## 6 Limitations

- **The planner is given the damaged plant exactly.** Every number here is an upper bound on what an FTC system achieves, because fault detection, isolation and identification are not modelled. In practice the plant estimate arrives late and wrong, and both shrink these numbers.
- **Open loop.** Each result is a single optimal-control solve, not a closed loop. There is no re-solve as the trajectory is flown, so a fault whose model error grows in flight (the drift case in particular) is treated more kindly here than a real one would be.
- **A box is not a distribution.** The regimes have uniform density and no covariance structure. They describe boxes, not the operational likelihood of being in them.
- **Dead time is quantised to the control grid.** A delay of one 1 s interval is exact on a zero-order-hold grid, but the framework's warning about 10–30 ms delays eroding phase margin belongs to a closed-loop bandwidth this 1 s planning grid cannot resolve.
- **Oscillation frequency is grid-limited.** Chugging is modelled at 0.25 Hz; the 50–500 Hz longitudinal modes and the kHz acoustic modes of framework §2.3 are far outside what a 1 s grid with 2 sub-steps can represent, and are a structural-dynamics problem rather than a trajectory one.
- **One fault at a time, on engine 2.** The vehicle is laterally symmetric so the engine index does not matter, but simultaneous or cascading faults are not covered.
- **Local infeasibility is not global.** The NLP is nonconvex; **no trajectory** means IPOPT failed from two independent seeds, not that none exists.
- **Constant mass,** inherited from the base model, which understates the propellant cost of a partial fault.
- **The easy regimes saturate.** In **On approach** and **Dispersed** the healthy control lands from nearly every admissible state, so those columns bound the fault effect from above rather than resolving it. The discrimination lives in the **Low and late** and **Critical** columns, and the pooled marginals inherit the mix.
- **Per-cell samples are small.** With 10 initial conditions per cell, an individual cell of the matrix carries a wide interval. The pooled marginals ( $n = 50$  per fault) and the paired comparisons are what the conclusions rest on.

## 7 Reproducing

```
python studies/fault_taxonomy/run_taxonomy_study.py 10 # the campaign
python studies/fault_taxonomy/analyse.py             # figures + headline JSON
python studies/fault_taxonomy/build_report.py         # this document
```